Md Zesun Ahmed Mia

R&D Engineer | Next-Generation Neuromorphic & Brain-Inspired Systems

Research engineer dedicated to building the next generation of brain-inspired computing hardware. Expertise in neuromorphic architectures, in-memory computing, and spintronic device integration.

Education

08/2022–08/2027 (Expected) **Ph.D. in Electrical Engineering**, *Pennsylvania State University*, University Park, PA, USA, *CGPA: 4.00/4.00*
Advisor: Dr. Abhronil Sengupta

08/2022–08/2024 **M.S. in Electrical Engineering**, *Pennsylvania State University*, University Park, PA, USA, *CGPA: 4.00/4.00*
Thesis: Neuromorphic Computing for Lifelong Learning

02/2015–04/2019 **B.Sc. in Electrical and Electronic Engineering**, *Bangladesh University of Engineering & Technology (BUET)*, Dhaka, Bangladesh, *CGPA: 3.81/4.00*

Industrial Experience

05/2025–07/2025 **Graduate Technical Intern**, *Intel Corporation*, Hillsboro, OR
- Applied AI and neural network approaches to semiconductor process optimization for neuromorphic and brain-inspired computing hardware.
- Developed machine learning models for process parameter prediction supporting AI accelerator and neuromorphic device fabrication.
- Designed DOE for thin film processes enabling enhanced performance in next-generation neuromorphic computing systems.

09/2020–07/2021 **R&D Engineer**, *SEMWAVES Ltd.*, London, UK, part-time
- Delivered a 50 kW hybrid renewable system (solar + smallscale hydro) for an offgrid Bangladeshi site, supporting reliable community power.
- Owned technical leadership and coordination (vendors, field teams, stakeholders); directed device selection, integration, QA/safety procedures, and optimization against load profiles and uptime targets.

Technical Skills

Neuromorphic	SNNs, STDP, event-driven processing, brain-inspired ML, Loihi 2
Devices, Memory & Fabrication	Spintronic NVM (MTJ, FeFET), ferroelectric/magnetic device fabrication, SRAM/DRAM, device-circuit co-design, thin film deposition (CVD, HDP-CVD, Sputtering), lithography, etching, electrical/optical/AFM/SEM characterization, magnetic probe station
HW-SW Co-Design	Algorithm-device-system integration, analog/mixed-signal circuits, hardware-in-the-loop training, PCB design, lab instrumentation, oscilloscope, signal generator, LabVIEW
EDA/Sim.	TCAD (Silvaco, Sentaurus), Cadence Virtuoso, Spectre, HSPICE, COMSOL, MATLAB, ModelSim, Synopsys, CIM (Compute-In-Memory) Systems
Programming & ML Tools	Python, MATLAB, Verilog, C/C++, Bash, Linux/Unix, PyTorch, TensorFlow, scikit-learn, Data Visualization, Pandas, NumPy, JMP, Jupyter, LaTeX
Collaboration	Git, Slack, Microsoft Office, Google Workspace
Agentic AI	Advanced use of generative and agentic AI tools (Cursor, Copilot, VSCode, Cline, ChatGPT Agents) for research, design, and code development

Academic Research and Teaching Experience

814-280-7244 • *zesun.ahmed@psu.edu* • *zesun33.github.io*
zesun-ahmed • *zesun33*

08/2022– Present **Graduate Research Assistant**, *Penn State*, University Park, PA
- Researched and implemented advanced brain-inspired learning algorithms, including Equilibrium Propagation, a novel Astromorphic Transformer, and RMAAT.
- Developed a hardware-aware training framework to optimize ML models for deployment on systems with integrated Non-Volatile Memory devices.
- Integrated fabricated spintronic devices into neuromorphic system simulations, using hardware-aware training to evaluate their potential as synaptic elements.
- Designed and evaluated SNN and DNN architectures for deployment on emerging hardware, focusing on energy efficiency and on-chip learning capabilities.

08/2024– 05/2025 **Graduate Teaching Assistant**, *Penn State*, State College, PA
- Taught Cadence Virtuoso (schematic/layout), PDK usage, DRC/LVS, and analog/digital design flows; created hands-on lab modules and guided tool/debug workflows.
- Supervised Capstone projects for 90+ undergraduate students across communications, electronics, and firmware: supported Raspberry Pi/Arduino development, software-defined radio experiments, PCB design, and system integration end-to-end.

02/2021– 08/2022 **Lecturer**, *University of Liberal Arts Bangladesh (ULAB)*, Dhaka, Bangladesh
- Taught undergraduate courses: Solid State Devices, Digital Circuit Design, Semiconductor Device Physics, Power Electronics.
- Developed lab modules and supervised student projects on semiconductor devices and circuits.

Projects

Astromorphic Transformer — Lead Student Researcher, 2022–2025. Developed a neuromorphic algorithmic framework for transformer models with astrocytic memory, enabling biologically inspired sequence learning. See publication: [IEEE TCDS].

Spintronic Device-Based Memory — Lead Student Researcher, 2023–Present. Led the fabrication and comprehensive characterization of spintronic memory arrays for machine learning systems, advancing non-volatile memory technologies to enhance computational performance.

Neuromorphic Cybersecurity — Lead Student Researcher, 2023–2025. Developed neuromorphic approaches for cybersecurity in lifelong learning systems. Accepted at ICONS 2025. See publication: [arXiv], [ICONS 2025].

Select Publications

[1] Md Zesun Ahmed Mia, Malyaban Bal, and Abhronil Sengupta. "Delving deeper into astromorphic transformers". In: *IEEE Transactions on Cognitive and Developmental Systems* (2025).

[2] Md Zesun Ahmed Mia et al. "Neuromorphic Cybersecurity with Semi-supervised Lifelong Learning". In: *arXiv preprint arXiv:2508.04610* (2025).

[3] Tao Zhang et al. "Self-sensitizable neuromorphic device based on adaptive hydrogen gradient". In: *Matter* 7.5 (2024), pp. 1799–1816.

[4] KM Ashraful Hoque Fahim et al. "Study of 3-nm Cylindrical GAAFETs with Variations in High-k Dielectric Gate-oxide Materials". In: *2022 IEEE Symposium on Industrial Electronics & Applications (ISIEA)*. IEEE. 2022, pp. 1–5.

Recognitions

- The Wormley Family Graduate Fellowship, Harry G. Miller Fellowships in Engineering (2025)
- Arthur Waynick Graduate Scholarship (2024)
- Milton and Albertha Langdon Memorial Fellowship (2023)
- Melvin P. Bloom Memorial Fellowship (2022)

Professional Affiliations

- Reviewer, Design Automation Conference (DAC) 2025, IEEE MWSCAS 2025, IACCESS 2024
- Student Member, IEEE (2015-Present), Executive Member, EDS, IEEE BD (2021-2022)

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